Cheng-Hao (Howard), Kao

About me

A second-year MASc student in Mechatronics engineering at the University of Waterloo. Proficient in machine learning and data analysis. Possess the analytical abilities required for research and design as well as a comprehensive grasp of engineering principles. Currently researching on multi-agent system workloads as an intern at Huawei, and working in the Structural Biomechanics Lab on ML-based injury prediction. Previously involved in the Robotics and Computer Vision Lab conducting research on multi-robot collaboration joint trajectory planning and control systems and in the Microelectromechanical Systems Lab dedicated to integration of FPCB mirror-based scanners. Seeking more professional training for further development within the field of AI, data analysis, and cognitive science.

Contact

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- Cheng-Hao Kao
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- Cheng-Hao Kao

Languages

Chinese - Native Language

English - Professional Knowledge

EDUCATION

January 2024 - Present

MASc in Mechanical and Mechatronics Engineering — Waterloo, On, Canada
University of Waterloo
GPA: 3.90/4.0 Cumulative

Relevant Coursework: Introduction to Machine Learning, Pattern Recognition and Machine Learning, Deployment of Deep Learning Model, Computational Intelligence, Optimization Methods, Human Factors in Testing

August 2018 - June 2023

Bachelor of Engineering — Toronto, On, Canada
Toronto Metropolitan University
GPA: 3.85/4.33 Cumulative

Relevant Coursework: Mechanical Design, Control of Robotic Manipulators, Real-Time Computer Control Systems, Fundamentals of Robotics, Mechatronics Systems Design, Measurements, Sensors and Instruments, Microprocessor Systems, Control Systems Digital Systems, Computer Programming Fundamentals, Electric Machines and Actuators, Electric Circuits, Applied Finite Elements, Vibrations, Machine Design, Mechanics of Machines, Solid Mechanics, Statics and Mechanics of Materials, Dynamics, Manufacturing Fundamentals, Introduction to Engineering Design, Engineering Graphical Communication, Numerical Analysis, Linear Algebra, Differential Equations and Vector Calculus, Calculus II, Calculus I

August 2015 - June 2018

High School Diploma — Shanghai, China
No.2 High School of East China Normal University
Average: 95.3/100

WORK EXPERIENCE

August 2025 - Present

Technology Planning and Cooperation Research Intern — Waterloo, On, Canada
Huawei Canada

- Multi-agent system workload trend identification
- Multi-agent system system-level optimization analysis.

May 2024 - April 2024

Teaching Assistant — Waterloo, On, Canada
University of Waterloo

- Teaching Circuits
- Teaching Mechanics of Deformable Solids
- Teaching Statics

June 2023 - January 2024

Research Assistant — Toronto, On, Canada
Toronto Metropolitan University RCVL Lab

- Research on multi-robot collaboration joint trajectory planning and control
- Research on micro-plastic detection via ultrasonic sensor and machine learning

Professional Skills

Deep Learning · Machine Learning · Data Analysis · Optimization · Signal Processing · Robotics · Control Systems · Microcontrollers · Simulation and Design · Engineering Mathematics

Honors and Awards

Dean's List 2018 to 2019 · Dean's List 2019 to 2020 · Dean's List 2020 to 2021 · Dean's List 2021 to 2022 · Mechanical Engineering Alumni Award · Mechanical Engineering Capstone Project Award · Graduation with Distinction

Soft Skills and Strengths

Adaptability · Problem Solving · Collaboration · Team Working · Open-Mindedness · Critical Thinking · Communication · Willingness to Learn · Responsibility · Reliability · Enthusiasm · Autonomy · Motivated · Stress Tolerance · Research and Analysing · Personal Time Management · Strategic Planning · Patience · Empathy · Discipline

Interests

- Deep Learning
- Machine Learning
- Cognitive Science
- Agentic AI
- Control Systems
- Robotics
- Card Games
- Bees
- Billiards
- Board Games
- Squash

September 2022 - February 2023

Research Assistant — Toronto, On, Canada
Toronto Metropolitan University MEMS Lab

- Research on FPCB mirror-based lidar scanners
- Integration of PSD sensors, ToF sensors, and GUI

April 2021 - September 2021

Research Assistant Intern — Toronto, On, Canada
Toronto Metropolitan University Ultrashort Laser Nanomanufacturing Lab

- Conducting porosity and fiber analyses via SEM image processing
- Generating molecular dynamics simulations

April 2020 - September 2020

Research Assistant Intern — Toronto, On, Canada
Toronto Metropolitan University Ultrashort Laser Nanomanufacturing Lab

- Conducting porosity and fiber analyses via SEM image processing
- Generating molecular dynamics simulations

PUBLICATIONS

Manuscript
2025
Predicting the Risk of ACL Injuries Using Machine Learning Models and Anatomical Predictors, *Under Review*, Link Unavailable

Arxiv Preprint
2024
Morphological Detection and Classification of Microplastics and Nanoplastics Emerged from Consumer Products by Deep Learning, *IEEE*, Link

Journal Article
2023
MR. CAP: Multi-Robot Joint Control and Planning for Object Transport, *IEEE Control System Letters*, Link

SOFTWARE SKILLS

Data Analysis	**PyTorch**	**TensorFlow**
	AutoGluon	**SkLearn**
	XGBoost	**CATBoost**
	SAM	**YOLO**
	NLopt	**Optuna**
Robotics	**GTSAM**	
Modeling and Simulation	**Simulink**	**Multisim**
	ANSYS	
Design and Development	**SolidWorks**	**Matlab**
	LabVIEW	
Version Control	**GitHub**	
Office Automation	**MS Office (Excel, Word, PowerPoint)**	
	Overleaf	**Adobe Illustrator**

</> PROGRAMMING LANGUAGES

- Python
- Matlab
- Processing
- C++
- C
- LaTex
- Fortran